Climate hazard at a facility

Reduced facility output

Reduced input availability downstream

Final product shortage

From localized climate risks to facility-level production losses

Climate hazards

- Heatwaves
- Tropical cyclones
- Wildfires

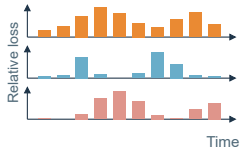
Global steel production facilities

2023 output · Mt Fe-equivalent



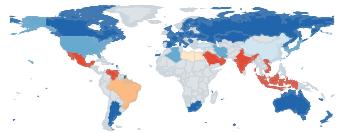
Stage	Example facility	Economy	Output
Iron mines (409)	Hamersley Consolidated	AUS	138.2
	Northern System	BRA	82.1
Ironmaking facilities (474)	POSCO Gwangyang	KOR	19.6
	Zhangjiagang Hongchang	CHN	16.9
Steel plants (871)	Angang	CHN	19.3
	POSCO Gwangyang	KOR	18.8

Facility-level production losses



Aggregate by country and production stage

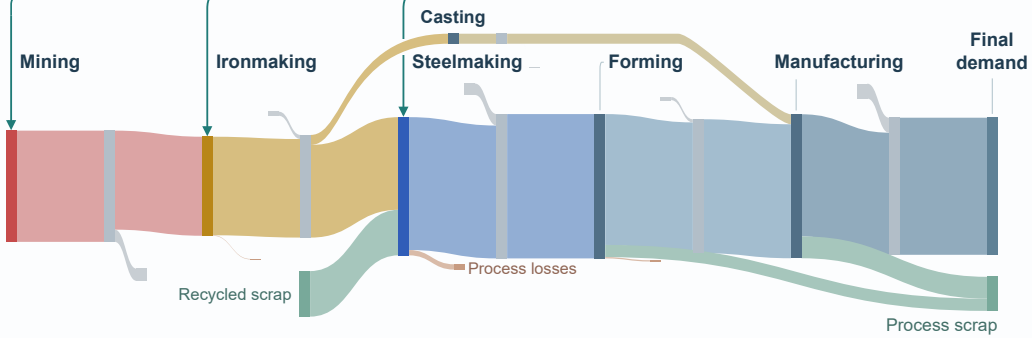
Country-stage production constraints



Upstream production capacity limits

From production losses to final steel supply

2023 steel material-flow network



Weekly dynamic simulation

Comparison settings

Capacity profile

P1 P2 P3 P4 P5 P6 P7

Uneven

Balanced

Climate pathways

Pre-industrial | SSP1-2.6 | SSP3-7.0 | SSP5-8.5

Inventory buffers (weeks)

1 2 3 4 5 6 7 8

Trade access

Open | Restricted

Production → Inventories → Trade matching → Final supply

Analysis of national steel security

Simulated supply outcomes

Supply maintained

With sufficient buffers and alternative supply

Supply shortfall

If constraints outlast available buffers

Domestic capability and final-demand satisfaction

Domestic supply chain capability index (DSCCI)

Baseline capacity adequacy and balance

Association

Final-demand satisfaction (%)

$$\frac{\text{Simulated final steel supply}}{\text{2023 baseline final demand}} \times 100$$